

PIZZA SALES DATABASE ANALYSIS

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PROJECT OVERVIEW

THIS PROJECT ANALYZES A REAL-WORLD PIZZA SALES DATABASE USING POSTGRESQL

.THE OBJECTIVE IS TO ANSWER BUSINESS QUESTIONS RELATED TO SALES PERFORMANCE, CUSTOMER ORDERING PATTERNS, REVENUE GENERATION, AND PRODUCT POPULARITY THROUGH SQL QUERIES.

THE PROJECT DEMONSTRATES PRACTICAL USE OF:

JOINS

AGGREGATE FUNCTIONS

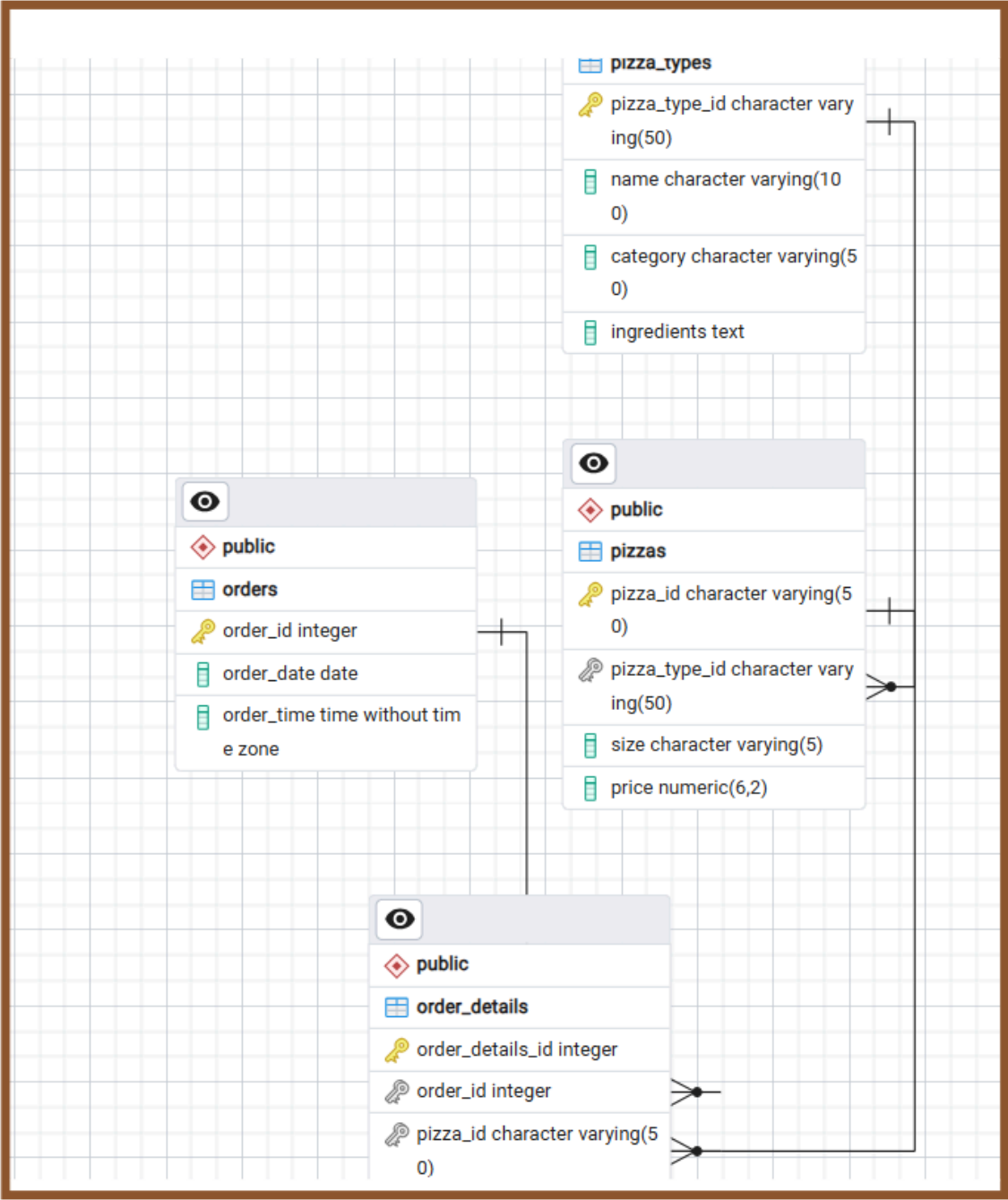
SUBQUERIES

CTES

WINDOW FUNCTIONS

RANKING FUNCTIONS

ERD DIAGRAM



1. RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED.

```
SELECT  
    COUNT(*) AS TOTAL_ORDER  
FROM  
    ORDERS;
```

	total_order bigint 
1	21350

2.CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES.

```
SELECT
    SUM(P.PRICE * O.QUANTITY) AS REVENUE
FROM
    PIZZAS P
    JOIN ORDER_DETAILS O ON P.PIZZA_ID = O.PIZZA_ID;
```

	revenue numeric 
1	817860.05

3.IDENTIFY THE HIGHEST-PRICED PIZZA.

```
SELECT
    PT.NAME,
    P.PRICE
FROM
    PIZZAS P
    JOIN PIZZA_TYPES PT ON P.PIZZA_TYPE_ID = PT.PIZZA_TYPE_ID
ORDER BY
    PRICE DESC
LIMIT
    1;
```

	name character varying (100) 	price numeric (6,2) 
1	The Greek Pizza	35.95

4.IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.

```
SELECT
    P.SIZE,
    COUNT(O.ORDER_DETAILS_ID) AS MOST_COMMON_SIZE
FROM
    ORDER_DETAILS O
    JOIN PIZZAS P ON P.PIZZA_ID = O.PIZZA_ID
GROUP BY
    P.SIZE
ORDER BY
    COUNT(O.ORDER_DETAILS_ID) DESC
LIMIT
    1;
```

	size character varying (5) 	most_common_size bigint 
1	L	18526

5.LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.

```
select * from order_details
SELECT
    PT.NAME,
    SUM(O.QUANTITY) AS NO_TIMES_ORDER
FROM
    ORDER_DETAILS O
    JOIN PIZZAS P ON P.PIZZA_ID = O.PIZZA_ID
    JOIN PIZZA_TYPES PT ON PT.PIZZA_TYPE_ID = P.PIZZA_TYPE_ID
GROUP BY
    PT.NAME
ORDER BY
    NO_TIMES_ORDER DESC
LIMIT
    5;
```

	name character varying (100)	no_times_order bigint
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Piz...	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

6. JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED.

```
select * from pizzas;
```

```
SELECT
    PT.CATEGORY,
    SUM(O.QUANTITY) AS TOTAL_QUANTITY
FROM
    ORDER_DETAILS O
    JOIN PIZZAS P ON O.PIZZA_ID = P.PIZZA_ID
    JOIN PIZZA_TYPES PT ON P.PIZZA_TYPE_ID = PT.PIZZA_TYPE_ID
GROUP BY
    PT.CATEGORY
ORDER BY
    TOTAL_QUANTITY DESC;
```

	category character varying (50) 	total_quantity bigint 
1	Classic	14888
2	Supreme	11987
3	Veggie	11649
4	Chicken	11050

7.DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.

```
SELECT
    EXTRACT(
        HOUR
        FROM
            ORDER_TIME
    ) AS HOUR_DISTRIBUTION,
    COUNT(ORDER_ID)
FROM
    ORDERS
GROUP BY
    HOUR_DISTRIBUTION;
```

8. JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS.

```
SELECT  
    CATEGORY,  
    COUNT (NAME)  
FROM  
    PIZZA_TYPES  
GROUP BY  
    CATEGORY;
```

	category character varying (50) 🔒	count bigint 🔒
1	Supreme	9
2	Chicken	6
3	Classic	8
4	Veggie	9

9.GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.

Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT
    AVG(TOTAL_ORDER_PIZZA) AS AVG_PIZZA_ORDER_PER_DAY
FROM
    (
        SELECT
            O.ORDER_DATE,
            SUM(OD.QUANTITY) AS TOTAL_ORDER_PIZZA
        FROM
            ORDERS O
            JOIN ORDER_DETAILS OD ON O.ORDER_ID = OD.ORDER_ID
        GROUP BY
            O.ORDER_DATE
    );
```

	avg_pizza_order_per_day 
1	138.4748603351955307

10.DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.

```
SELECT
    PT.NAME,
    SUM(P.PRICE * OD.QUANTITY) AS REVENUE
FROM
    ORDER_DETAILS OD
    JOIN PIZZAS P ON P.PIZZA_ID = OD.PIZZA_ID
    JOIN PIZZA_TYPES PT ON PT.PIZZA_TYPE_ID = P.PIZZA_TYPE_ID
GROUP BY
    PT.NAME
ORDER BY
    REVENUE DESC
LIMIT
    3;
```

	name  character varying (100)	revenue  numeric
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Piz...	42768.00
3	The California Chicken Piz...	41409.50

11.CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.

```
select distinct pizza_type_id from pizza_types;

WITH
    REVENUE AS (
        SELECT
            P.PIZZA_TYPE_ID,
            SUM(P.PRICE * OD.QUANTITY) AS TOTAL_REVENUE
        FROM
            PIZZAS P
            JOIN ORDER_DETAILS OD ON P.PIZZA_ID = OD.PIZZA_ID
        GROUP BY
            P.PIZZA_TYPE_ID
    )
SELECT
    PIZZA_TYPE_ID,
    ROUND(
        TOTAL_REVENUE * 100 / (
            SELECT
                SUM(TOTAL_REVENUE)
            FROM
                REVENUE
        ),
        2
    ) AS PER_REVE
FROM
    REVENUE
```

	pizza_type_id character varying (50) 🔒	per_reve numeric 🔒
1	bbq_ckn	5.23
2	pep_msh_pep	2.30
3	mediterraneo	1.88
4	mexicana	3.27
5	ital_supr	4.09
6	big_meat	2.81
7	ital_veggie	1.96
8	napolitana	2.95
9	hawaiian	2.05

12.ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.

```
SELECT
ORDER_DATE,
REVENUE AS PRE_REVENUE,
SUM(REVENUE) OVER (
ORDER BY
ORDER_DATE
) AS CUMMULATIVE
FROM
(
SELECT
O.ORDER_DATE,
SUM(PRICE * QUANTITY) AS REVENUE
FROM
ORDERS O
JOIN ORDER_DETAILS OD ON O.ORDER_ID = OD.ORDER_ID
JOIN PIZZAS P ON P.PIZZA_ID = OD.PIZZA_ID
GROUP BY
O.ORDER_DATE
)
```

	order_date 	pre_revenue 	cummulative 
1	2015-01-01	2713.85	2713.85
2	2015-01-02	2731.90	5445.75
3	2015-01-03	2662.40	8108.15
4	2015-01-04	1755.45	9863.60
5	2015-01-05	2065.95	11929.55
6	2015-01-06	2428.95	14358.50
7	2015-01-07	2202.20	16560.70
8	2015-01-08	2838.35	19399.05
9	2015-01-09	2127.35	21526.40
10	2015-01-10	2162.25	23688.65

13.DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.

```
select category,name,revenue,rank
From
(select category,name,revenue,
DENSE_RANK() over(partition by category order by revenue desc ) as rank
from
(SELECT
    PT.CATEGORY,pt.name,
    SUM(P.PRICE * OD.QUANTITY) AS REVENUE
FROM
    ORDER_DETAILS OD
    JOIN PIZZAS P ON P.PIZZA_ID = OD.PIZZA_ID
    JOIN PIZZA_TYPES PT ON PT.PIZZA_TYPE_ID = P.PIZZA_TYPE_ID
GROUP BY
    PT.CATEGORY,pt.pizza_type_id) as revenue)
where rank<=3;
```

	category character varying (50) 🔒	name character varying (100) 🔒	revenue numeric 🔒	rank bigint 🔒
1	Chicken	The Thai Chicken Pizza	43434.25	1
2	Chicken	The Barbecue Chicken Piz...	42768.00	2
3	Chicken	The California Chicken Piz...	41409.50	3
4	Classic	The Classic Deluxe Pizza	38180.50	1
5	Classic	The Hawaiian Pizza	32273.25	2
6	Classic	The Pepperoni Pizza	30161.75	3
7	Supreme	The Spicy Italian Pizza	34831.25	1
8	Supreme	The Italian Supreme Pizza	33476.75	2
9	Supreme	The Sicilian Pizza	30940.50	3

KEY BUSINESS INSIGHTS

 **PEAK ORDER HOURS IDENTIFIED USING HOURLY ORDER ANALYSIS.**

 **REVENUE CONTRIBUTION OF EACH PIZZA CATEGORY WAS CALCULATED.**

 **TOP-PERFORMING PIZZA TYPES WERE IDENTIFIED USING WINDOW FUNCTIONS.**

 **HIGHEST REVENUE-GENERATING PIZZAS WERE RANKED USING DENSE_RANK().**

 **DAILY SALES TRENDS AND CUMULATIVE REVENUE WERE ANALYZED.**

 **COMPLEX SQL CONCEPTS SUCH AS CTES, JOINS, AGGREGATE FUNCTIONS, AND WINDOW FUNCTIONS WERE APPLIED TO SOLVE REAL-WORLD BUSINESS PROBLEMS.**